Manda Ramya

Coffee Shop Sales Analysis Report

1.Dataset Description

The Coffee Shop Sales dataset captures detailed transaction-level data from multiple coffee shop outlets. Each row represents a single sale, enabling comprehensive analysis of sales patterns, product performance, and customer demand.

It contains 11 columns:

- transaction_id: Unique identifier for each sale.
- transaction_date: Date of transaction.
- transaction_time: Time of purchase.
- transaction_qty: Quantity of items purchased.
- store_id: Unique ID of each store.
- store_location: Location of the store (e.g., Lower Manhattan, Astoria, Hell's Kitchen).
- product_id: Unique ID for each product.
- unit_price: Price per unit of the item.
- product_category: Broad category of the item (e.g., Coffee, Tea, Bakery).
- product_type: Sub-type within the product category

2.Operations Performed

1. **Data Exploration and Schema Analysis**
   The dataset was loaded and inspected to understand its structure. The schema, column types, and sample records were reviewed to verify data integrity. Summary statistics were computed to check for missing values and outliers.
2. **Store and Location Analysis**
   The dataset was grouped by store_location to evaluate which outlets performed best in terms of total transactions and sales. It was found that **Lower Manhattan** had the highest number of sales, making it the top-performing branch.
3. **Product Category Evaluation**
   Sales were aggregated by product_category and product_type to identify top-selling

items. Coffee-based beverages emerged as the leading product line, followed by Tea and Bakery items. This step helped understand customer preferences.

4. **Revenue Computation**
   A new column was derived to calculate **total revenue per transaction** (transaction_qty * unit_price). Then, total revenue per store and category was analyzed to compare profitability across outlets and product lines.

5. **Time-Based Sales Analysis**
   Sales trends were analyzed by transaction_date and transaction_time to identify high-demand hours and days. This revealed peak sales during morning hours and weekends, useful for staffing and marketing decisions.

6. **Monthly and Weekly Trends**
   The dataset was grouped by month and day of the week to track periodic patterns. Seasonal changes in coffee sales were observed, especially in colder months when demand for hot drinks increased.

7. **Customer Behavior Insights**
   Transaction frequency and basket size (average quantity purchased) were analyzed to understand buying habits. It showed that most customers purchased single items, but bundle purchases increased during promotional periods.

8. **Price and Quantity Correlation**
   The relationship between unit_price and transaction_qty was checked to see if pricing affected purchase volume. Moderate price elasticity was observed, suggesting slight sensitivity to price changes.

9. **Filtering and Sorting Operations**
   Specific subsets of the data (e.g., transactions above certain revenue or products with high sales) were extracted to study targeted segments and outliers.

10. **Data Cleaning and Transformation**
    Columns were renamed, unnecessary fields dropped, and data types adjusted for consistent analysis using PySpark DataFrame transformations.

3.Key Visualizations (5 Major Plots)

1. **Bar Chart – Store-wise Total Sales**
   Compared revenue across all store locations, highlighting Lower Manhattan as the leading outlet.

2. **Pie Chart – Product Category Distribution**
   Illustrated sales proportions across Coffee, Tea, Chocolate, and Bakery items.

3. **Line Chart – Daily Sales Trend**
   Displayed fluctuations in daily revenue over time, revealing weekday and weekend differences.

4. **Bar Chart – Monthly Sales Trend**
   Showed monthly variations in sales, peaking during colder months.
5. **Heatmap – Hourly vs. Day-of-Week Sales**
   Visualized the busiest hours and days, confirming strong morning rush sales patterns.

4. Summary of operations performed

- Read CSV into DataFrame
- Visualization (plots/graphs)
- Summary statistics & frequency counts
- Grouping and aggregation (groupby)
- Sorting data
- Datetime parsing and feature extraction

5. Key Insights:

Weekly Sales Patterns: Sales vary across different days of the week, helping identify peak business days and low-traffic periods. This insight can guide staffing and operational planning.

Revenue Variability: The dataset shows the minimum, maximum, and total sales per day, highlighting days with exceptionally high or low revenue, which can inform targeted promotions.

Product Contribution: Different products consistently contribute to overall revenue, indicating balanced demand and potential for focused inventory management.

Store Performance: Multiple store locations show steady contributions to total sales, suggesting consistent operational efficiency across branches.

Strategic Planning Potential: The dataset supports trend analysis, inventory optimization, and promotional strategy development to improve revenue, reduce waste, and enhance operational decision-making.

6. **Recommendations**

1. **Focus Marketing on High-Demand Hours and Days**
   Increase staffing, offers, and targeted promotions during peak morning hours and weekends to maximize revenue during high-traffic times.
2. **Expand Coffee Product Range**
   Since coffee dominates sales, introducing new flavors, seasonal blends, and combo offers could attract repeat customers and increase basket size.

3. **Boost Underperforming Locations**
   Locations like Astoria and Hell's Kitchen can benefit from local promotions, discounts, or loyalty programs to improve footfall.
4. **Promote Complementary Items**
   Bundle coffee with bakery items (e.g., croissant + cappuccino offers) to raise average order value and encourage multiple-item purchases.
5. **Leverage Seasonal Demand**
   Offer limited-time drinks (e.g., winter specials or summer cold brews) to capitalize on seasonal preferences and keep the menu dynamic.
6. **Optimize Inventory by Time Trends**
   Use time-based sales insights to ensure that high-demand products are adequately stocked during peak hours and months, reducing waste and shortages.
7. **Implement Price Optimization**
   Since demand is not highly price-sensitive, moderate price increases for premium items could improve margins without significantly affecting volume.
8. **Introduce Loyalty and Subscription Programs**
   Regular customers could be incentivized with reward points, prepaid coffee subscriptions, or discounts on recurring purchases.

7. Conclusion

The dataset provides insights into the sales patterns of a mid-sized coffee shop chain across multiple stores and products. Day-wise analysis shows how sales fluctuate throughout the week, helping identify peak business days. The minimum, maximum, and total sales per day reveal the variability in daily revenue, highlighting high-performing days and potential opportunities to boost lower-performing periods. Products and store locations show consistent contribution to overall revenue, suggesting balanced demand across offerings. This dataset is well-suited for sales trend analysis, inventory planning, and promotional strategy development to optimize revenue and operational efficiency